

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \quad a > 0 \\ a \cdot c - b^2 > 0 \quad \langle w, v \rangle = w^T A v$$

$\langle w, v \rangle = \langle v, w \rangle$  *poti, matrica je realna i simetrična!*

$$\langle w, w \rangle \geq 0 \quad \forall w \in V;$$

$$w = (x, y)^T$$

$$\langle w, w \rangle = w^T A w = a x^2 + 2b xy + c y^2 =$$

$$= a \cdot \left( x^2 + 2 \frac{b}{a} xy \right) + c y^2 =$$

$$= a \cdot \left( x + \frac{b}{a} y \right)^2 - \frac{b^2}{a} y^2 + c y^2 =$$

$$= a \left( x + \frac{b}{a} y \right)^2 + \left( c - \frac{b^2}{a} \right) y^2 =$$

$$= a \left( x + \frac{b}{a} y \right)^2 + \frac{a \cdot c - b^2}{a} y^2 =$$

$$\stackrel{\geq}{=} \underbrace{a}_{>0} \underbrace{\left( x + \frac{b}{a} y \right)^2}_{\geq 0} + \underbrace{\frac{\det A}{a}}_{>0} \underbrace{y^2}_{\geq 0} \geq 0$$

$$\langle w, w \rangle = 0 \Leftrightarrow w = \vec{0} \quad \text{jerne iz } \det A \neq 0 \\ A \text{ je regularna!}$$